

LINEAR PREDICTION FOR SYSTEM IDENTIFICATION

A Project Report

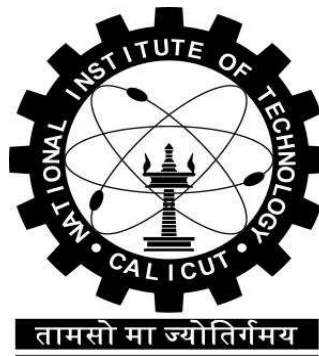
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*On the project work done as part of the Internship Program in the Department of
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CERTIFICATE

This is to certify that this report titled **Linear Prediction for System Identification** is a bonafide record of the project work done by **Jerome Punnoose**, APJ Abdul Kalam Technological University, MGP24ECE033, Electronics and Communication Engineering, Saintgits College of Engineering Kottayam as part of the internship program in the Department of Electronics and Communication Engineering, National Institute of Technology Calicut from 25th May 2026 to 23rd June 2026.

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Abstract

Linear Prediction (LP) is a fundamental technique in digital signal processing that models a signal as a linear combination of its past samples. By minimizing the mean-square prediction error, the optimal predictor coefficients can be determined and used to characterize the underlying dynamics of the signal source. Owing to its efficiency and accuracy, linear prediction has found extensive applications in speech processing, system identification, spectral estimation, and signal compression.

In this work, an unknown linear system is identified using the principles of linear prediction. The autocorrelation method is employed to estimate the signal statistics, and the resulting Yule–Walker equations are solved to obtain the optimal predictor coefficients. The Levinson recursion algorithm is utilized to provide a computationally efficient solution for determining the predictor parameters while ensuring numerical stability. The obtained coefficients are subsequently used to construct the prediction-error filter and the corresponding autoregressive (AR) model representation of the system.

Furthermore, the predictor coefficients are transformed into reflection coefficients through Levinson recursion, enabling the realization of both analysis and synthesis lattice filters. The lattice structure provides an efficient implementation of the prediction process and offers insight into the progressive decorrelation and reconstruction properties of the signal. Correlation characteristics at the outputs of successive lattice stages are investigated to demonstrate the whitening effect of the analysis filter and the signal reconstruction capability of the synthesis filter.

The performance of the identified model is validated through MATLAB/Python-based simulations. Prediction error analysis, autocorrelation studies, and signal reconstruction results are presented to verify the accuracy of the estimated model. The obtained results confirm that linear prediction provides an effective framework for system identification and signal modeling while maintaining computational efficiency and robust numerical behavior.

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1 Introduction

1.1 Background

System identification is the process of developing a mathematical model of an unknown dynamic system using observed data. Such models are essential in signal processing, communications, control systems, and speech analysis, where understanding the behavior of a system is necessary for analysis, prediction, and simulation. In many practical situations, only the output signal of the system is available, making it necessary to infer the system characteristics directly from the observed data.

Linear prediction is a widely used technique for system identification and signal modeling. It is based on the assumption that the current sample of a signal can be approximated as a linear combination of its past samples. That is, the signal has high sample to sample correlation. The predictor coefficients are determined by minimizing the mean-square prediction error, resulting in an optimal representation of the signal in the least-squares sense. These coefficients provide valuable information about the underlying system and can be used to construct an autoregressive (AR) model.

The determination of the predictor coefficients relies on the autocorrelation properties of the signal. By applying the orthogonality principle which is a consequence of optimality, the prediction problem leads to the Yule–Walker equations, which relate the predictor coefficients to the autocorrelation sequence. These equations can be solved efficiently using the Levinson recursion algorithm, which reduces computational complexity and simultaneously generates the reflection coefficients required for lattice filter realization. [1]

Lattice filters offer a numerically stable and modular implementation of the prediction process. The reflection coefficients obtained from Levinson recursion serve as the model parameters of the lattice structure. The analysis lattice filter progressively decorrelates the input signal, while the synthesis lattice filter reconstructs the original signal from the prediction residual.

In this project, linear prediction is employed to identify the characteristics of an unknown system using a speech signal. The Yule–Walker equations and Levinson recursion algorithm are used to obtain the predictor and reflection coefficients. The identified model is realized using analysis and synthesis lattice filters, and its performance is evaluated through simulation studies involving prediction-error analysis, autocorrelation measurements, and signal reconstruction.

1.2 Problem Statement

Many practical systems can be characterized only through observed output signals, while their internal structure and parameters remain unknown. Determining an accurate mathematical model of such systems is essential for analysis, prediction, and signal reconstruction. Conventional modeling approaches often require prior knowledge of the system dynamics or involve computationally intensive estimation techniques.

The problem addressed in this project is the identification of an unknown linear system from a speech signal using linear prediction techniques. The objective is to estimate the system model by minimizing the mean-square prediction error and determining the optimal predictor coefficients. Furthermore, the predictor coefficients are to be transformed into reflection coefficients using Levinson recursion and employed in the realization of analysis and synthesis lattice filters.

The effectiveness of the identified model is evaluated through prediction-error analysis, autocorrelation studies, gapped-function analysis, and signal reconstruction. The study aims to demonstrate that linear prediction and lattice filter structures provide an efficient and accurate framework for system identification and signal modeling.

2 Theory of Linear Prediction [1]

Linear prediction is a signal modeling technique in which the current sample of a signal is estimated as a linear combination of its past samples. This method is possible only if there is high sample to sample correlation. The objective is to determine a set of predictor coefficients that minimizes the prediction error in the mean-square sense. Linear prediction has extensive applications in speech processing, spectral estimation, system identification, and data compression.

2.1 Linear Prediction Model

$$y(n) = a_1y(n-1) + a_2y(n-2) + \cdots + a_py(n-p) \quad (2.1)$$

where $a_1, a_2, \dots, a_p$ denote the prediction coefficients and p represents the order of the predictor.

The difference between the actual sample and its predicted value is known as the prediction error. Thus,

$$e(n) = y(n) - \hat{y}(n) \quad (2.2)$$

For a second-order predictor,

$$e(n) = y(n) - a_1y(n-1) - a_2y(n-2) \quad (2.3)$$

The objective of linear prediction is to determine the coefficients (a_k) such that the mean-square value of the prediction error is minimized. The prediction error power is defined as

$$E = E[e^2(n)] \quad (2.4)$$

where ($E[\cdot]$) denotes the expectation operator.

According to the orthogonality principle, the optimum prediction error must be orthogonal to all past samples used in the prediction process. Therefore,

$$E[e(n)y(n-m)] = 0 \quad (2.5)$$

for

$$m = 1, 2, \dots, p. \quad (2.6)$$

Substituting the expression for the prediction error,

$$E \left[\left(y(n) - \sum_{k=1}^p a_k y(n-k) \right) y(n-m) \right] = 0 \quad (2.7)$$

which gives

$$r(m) = \sum_{k=1}^p a_k r(m-k) \quad (2.8)$$

Equation (2.8) represents a set of linear equations known as the Yule–Walker equations. For a predictor of order p , these equations can be written as

$$r(1) = a_1 r(0) + a_2 r(1) + \cdots + a_p r(p-1) \quad (2.9)$$

$$r(2) = a_1 r(1) + a_2 r(0) + \cdots + a_p r(p-2) \quad (2.10)$$

$\vdots$

$$r(p) = a_1 r(p-1) + a_2 r(p-2) + \cdots + a_p r(0) \quad (2.11)$$

The above equations may be expressed in matrix form as

$$\begin{bmatrix} r(0) & r(1) & r(2) & \cdots & r(p-1) \\ r(1) & r(0) & r(1) & \cdots & r(p-2) \\ r(2) & r(1) & r(0) & \cdots & r(p-3) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ r(p-1) & r(p-2) & r(p-3) & \cdots & r(0) \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_p \end{bmatrix} = \begin{bmatrix} r(1) \\ r(2) \\ \vdots \\ r(p) \end{bmatrix} \quad (2.12)$$

or compactly as

$$\mathbf{R}\mathbf{a} = \mathbf{r} \quad (2.13)$$

where $\mathbf{R}$ is the autocorrelation matrix, $\mathbf{a}$ is the predictor coefficient vector, and $\mathbf{r}$ is the autocorrelation vector.

The solution of the Yule–Walker equations yields the optimal predictor coefficients in the minimum mean-square error sense. These coefficients characterize the correlation structure of the signal and form the basis for system identification. In this work, the predictor coefficients are determined using the Levinson recursion algorithm, which exploits the Toeplitz structure of the autocorrelation matrix to obtain an efficient recursive solution.

2.2 Formation of the Autocorrelation Matrix

The calculated autocorrelation values are used to construct the autocorrelation matrix

$$\mathbf{R} = \begin{bmatrix} r(0) & r(1) & r(2) & \cdots & r(p-1) \\ r(1) & r(0) & r(1) & \cdots & r(p-2) \\ r(2) & r(1) & r(0) & \cdots & r(p-3) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ r(p-1) & r(p-2) & r(p-3) & \cdots & r(0) \end{bmatrix} \quad (2.14)$$

The matrix $\mathbf{R}$ is symmetric and possesses a Toeplitz structure, meaning that its elements are constant along each diagonal. This property significantly reduces the computational complexity associated with solving the Yule–Walker equations.

3 Levinson Recursion Algorithm [1]

3.1 Introduction

The predictor coefficients obtained from the Yule–Walker equations can be determined either by direct matrix inversion or by recursive techniques. Since the autocorrelation matrix possesses a symmetric Toeplitz structure, the Levinson recursion algorithm provides a computationally efficient method for solving the Yule–Walker equations. In addition to estimating the predictor coefficients, the algorithm also generates the reflection coefficients required for lattice filter realization.

In this work, Levinson recursion is employed to compute both the predictor coefficients and the reflection coefficients from the estimated autocorrelation sequence.

3.2 Need for Levinson Recursion

Direct inversion of the autocorrelation matrix requires a large number of computations, particularly for higher predictor orders. The computational complexity of direct matrix inversion is approximately

$$O(p^3) \quad (3.1)$$

where p is the predictor order.

Since the autocorrelation matrix is Toeplitz, Levinson recursion exploits this structure to reduce the computational complexity to

$$O(p^2). \quad (3.2)$$

Consequently, Levinson recursion is widely used in linear prediction, speech processing, and system identification applications.

3.3 Reflection Coefficients

The Levinson recursion algorithm determines the predictor coefficients incrementally. At each stage, a new parameter known as the reflection coefficient is computed.

For an m^{th} -order predictor, the reflection coefficient is given by

$$k_m = -\frac{r(m) + \sum_{i=1}^{m-1} a_i^{(m-1)} r(m-i)}{E_{m-1}} \quad (3.3)$$

where

- k_m is the reflection coefficient at stage m ,
- $a_i^{(m-1)}$ are the predictor coefficients of order $m - 1$,
- E_{m-1} is the prediction error energy of the previous stage.

The reflection coefficient represents the residual correlation that remains after the lower-order predictor has been applied.

3.4 Predictor Coefficient Update

After computing the reflection coefficient, the predictor coefficients are updated recursively according to

$$a_i^{(m)} = a_i^{(m-1)} + k_m a_{m-i}^{(m-1)} \quad (3.4)$$

for

$$i = 1, 2, \dots, m - 1 \quad (3.5)$$

and

$$a_m^{(m)} = k_m. \quad (3.6)$$

Thus, the predictor coefficients of a higher-order model are obtained directly from those of the lower-order model.

3.5 Prediction Error Energy Update

The minimum prediction-error energy is updated at each stage as

$$E_m = E_{m-1}(1 - k_m^2) \quad (3.7)$$

with the initial condition

$$E_0 = r(0). \quad (3.8)$$

Since

$$|k_m| < 1 \quad (3.9)$$

for a stable predictor, the prediction-error energy decreases as the predictor order increases.

3.6 Levinson Recursion Procedure

The complete Levinson recursion algorithm may be summarized as follows:

1. Compute the autocorrelation sequence $r(k)$.
2. Set the initial prediction error energy as

$$E_0 = r(0).$$

3. For $m = 1, 2, \dots, p$:
 - (a) Compute the reflection coefficient k_m .
 - (b) Update the predictor coefficients.
 - (c) Update the prediction-error energy.
4. Continue the recursion until the desired predictor order is reached.

At the end of the recursion process, the optimal predictor coefficients and reflection coefficients are obtained.

4 Analysis and Synthesis Lattice Filter Realization [1]

4.1 Introduction

The predictor coefficients obtained through Levinson recursion may be realized either in direct form or lattice form. In the present work, a lattice realization is adopted because of its modular structure, numerical stability, and efficient implementation. The lattice filter is completely characterized by the reflection coefficients generated by the Levinson recursion algorithm.

A lattice filter consists of a cascade of identical stages. Each stage performs a decorrelation operation using a reflection coefficient and generates forward and backward prediction errors. The analysis lattice filter transforms a correlated signal into a less correlated residual sequence, while the synthesis lattice filter performs the inverse operation and reconstructs the original signal from the residual.

4.2 Forward and Backward Prediction Errors [1]

The operation of a lattice filter is based on two signals known as the forward prediction error and the backward prediction error.

The forward prediction error at stage m is denoted by

$$e_m^+(n) \quad (4.1)$$

and represents the error obtained while predicting the present sample from past samples.

Similarly, the backward prediction error is denoted by

$$e_m^-(n) \quad (4.2)$$

and represents the error obtained while predicting past samples from future samples.

At the input of the lattice structure,

$$e_0^+(n) = y(n) \quad (4.3)$$

and

$$e_0^-(n) = y(n). \quad (4.4)$$

These signals form the initial conditions for the recursive lattice computations.

4.3 Analysis Lattice Filter

The analysis lattice filter generates the forward and backward prediction errors recursively using the reflection coefficients obtained from Levinson recursion.

For the m^{th} stage,

$$e_m^+(n) = e_{m-1}^+(n) + k_m e_{m-1}^-(n-1) \quad (4.5)$$

and

$$e_m^-(n) = k_m e_{m-1}^+(n) + e_{m-1}^-(n-1) \quad (4.6)$$

where k_m is the reflection coefficient associated with the m^{th} lattice stage.

These equations are evaluated successively for all stages of the lattice structure. The final forward prediction error

$$e_p^+(n) \quad (4.7)$$

represents the residual signal or prediction-error sequence.

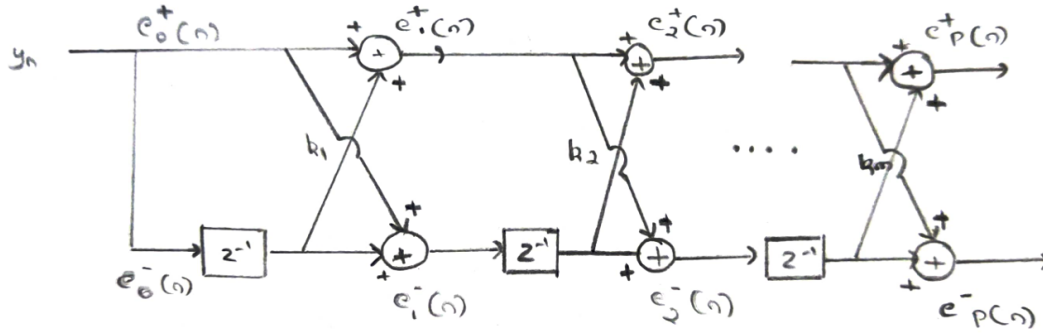


Figure 4.1: Analysis Lattice Filter

4.4 Gapped Functions

The intermediate forward and backward prediction errors generated at each lattice stage with the autocorrelation function as input are commonly referred to as gapped functions.

The forward gapped functions are

$$e_1^+(n), e_2^+(n), \dots, e_p^+(n) \quad (4.8)$$

while the backward gapped functions are

$$e_1^-(n), e_2^-(n), \dots, e_p^-(n). \quad (4.9)$$

These functions provide insight into the progressive decorrelation of the signal within the lattice structure. As the stage number increases, the forward prediction errors become increasingly uncorrelated, indicating successful removal of predictable information from the signal.

In this work, the gapped functions obtained at various lattice stages are analyzed to study the decorrelation properties of the prediction process.

4.5 Synthesis Lattice Filter

The synthesis lattice filter reconstructs the original signal from the residual sequence generated by the analysis lattice filter.

The synthesis process is performed using the same reflection coefficients employed in the analysis lattice. For the m^{th} stage,

$$f_m(n) = f_{m-1}(n) - k_m b_{m-1}(n-1) \quad (4.10)$$

and

$$b_m(n) = b_{m-1}(n-1) + k_m f_{m-1}(n) \quad (4.11)$$

where

- $f_m(n)$ denotes the forward synthesis signal,
- $b_m(n)$ denotes the backward synthesis signal,
- k_m is the reflection coefficient.

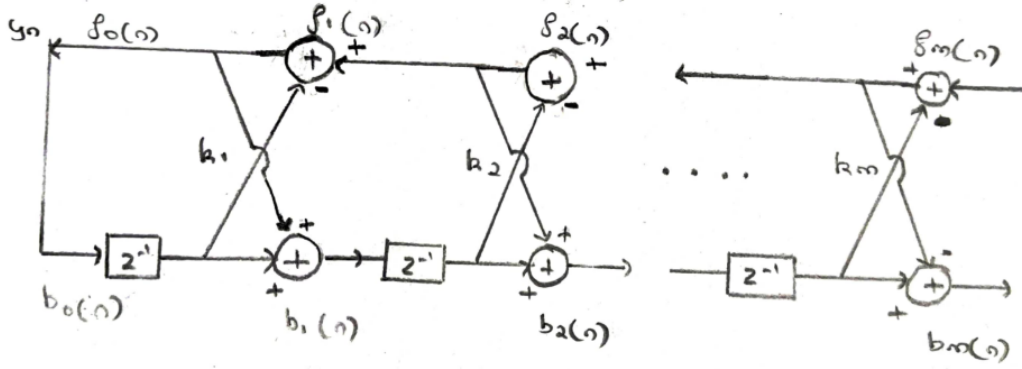


Figure 4.2: Synthesis Lattice Filter

The synthesis stages are connected in cascade to reconstruct the original speech signal from the residual sequence.

4.6 Reflection Coefficients as Model Parameters

The lattice realization is completely specified by its reflection coefficients

$$k_1, k_2, \dots, k_p. \quad (4.12)$$

Unlike the direct-form realization, where the predictor coefficients constitute the model parameters, the reflection coefficients directly characterize each stage of the lattice structure.

Since the analysis and synthesis filters used in this work are implemented in lattice form, the reflection coefficients obtained through Levinson recursion represent the parameters of the identified system. Therefore, these coefficients are reported as the optimal model parameters in the results section.

5 System Identification Using Linear Prediction

5.1 Introduction

System identification involves the determination of a mathematical model that accurately represents the behavior of an unknown system. In linear prediction, the predictor coefficients obtained from the observed signal provide information about the dynamics of the system that generated the signal. Consequently, the estimation of predictor coefficients can be viewed as a system identification problem.

In the present work, the unknown system is modeled as an autoregressive (AR) process. The parameters of the model are estimated using the autocorrelation method and Levinson recursion. The resulting model is then realized using lattice filter structures.

5.2 Autoregressive System Model

An autoregressive (AR) model represents the current sample of a signal as a linear combination of its previous samples and an excitation signal. The AR model of order p is given by

$$y(n) = a_1y(n-1) + a_2y(n-2) + \cdots + a_py(n-p) + e(n) \quad (5.1)$$

where

- $y(n)$ is the output signal,
- $a_1, a_2, \dots, a_p$ are the model coefficients,
- $e(n)$ is the excitation or prediction-error sequence.

The predictor coefficients characterize the dynamic behavior of the system and therefore serve as the basis for system identification.

5.3 Prediction Error Filter

Rearranging the autoregressive model yields

$$e(n) = y(n) - a_1y(n-1) - a_2y(n-2) - \cdots - a_py(n-p) \quad (5.2)$$

Taking the z -transform,

$$E(z) = A(z)Y(z) \quad (5.3)$$

where

$$A(z) = 1 - a_1z^{-1} - a_2z^{-2} - \dots - a_pz^{-p}. \quad (5.4)$$

The transfer function $A(z)$ is known as the prediction-error filter. When applied to the signal, it removes the predictable component and produces the residual sequence.

5.4 Identified System Transfer Function

The inverse of the prediction-error filter represents the identified system model. Therefore,

$$H(z) = \frac{Y(z)}{E(z)} = \frac{1}{A(z)} \quad (5.5)$$

or

$$H(z) = \frac{1}{1 - a_1z^{-1} - a_2z^{-2} - \dots - a_pz^{-p}}. \quad (5.6)$$

This transfer function constitutes the mathematical model of the identified system.

5.5 Pole Representation

The poles of the identified system are obtained from the roots of the denominator polynomial from equation (5.5)

$$A(z) = 1 - a_1z^{-1} - a_2z^{-2} - \dots - a_pz^{-p}. \quad (5.7)$$

The pole locations determine the spectral characteristics, stability, and dynamic response of the system. Hence, the predictor coefficients provide a complete description of the identified autoregressive model.

5.6 Lattice Representation of the Identified System

Although the identified system may be represented using the direct-form transfer function $H(z)$, the practical realization adopted in this work is based on lattice filters.

The reflection coefficients

$$k_1, k_2, \dots, k_p \quad (5.8)$$

obtained from Levinson recursion completely specify the lattice realization of the system. Each reflection coefficient corresponds to a stage of the lattice structure and represents the residual correlation present at that stage.

Therefore, in the lattice realization, the reflection coefficients constitute the model parameters of the identified system.

6 Simulation Methodology and Implementation

6.1 Introduction

The theoretical concepts of linear prediction, Levinson recursion, and lattice filter realization were implemented using Python. The simulation was carried out to estimate the predictor coefficients, obtain the reflection coefficients, realize the analysis and synthesis lattice filters, and verify the identified system through signal reconstruction.

The implementation consists of frame-based processing, autocorrelation estimation, predictor coefficient computation, lattice filter realization, and performance evaluation.

6.2 Speech Data

A sampled speech signal was used as the input for the system identification process. The speech samples were represented as a discrete-time sequence and processed frame by frame.

The sampling frequency was chosen as

$$F_s = 8000 \text{ Hz} \quad (6.1)$$

which corresponds to a sampling interval of

$$T_s = \frac{1}{F_s} = 125 \mu s. \quad (6.2)$$

The speech signal served as the observed output sequence from which the system parameters were estimated.

6.3 Frame-Based Processing

The speech signal was divided into short-duration frames before analysis. A frame duration of

$$T_f = 4 \text{ ms} \quad (6.3)$$

was selected.

The corresponding frame length was computed as

$$N = F_s T_f \quad (6.4)$$

which gives

$$N = 8000 \times 0.004 = 32 \quad (6.5)$$

samples per frame.

Each frame was processed independently to obtain the predictor coefficients and reflection coefficients associated with that segment of speech.

6.4 Autocorrelation Computation

For every speech frame, the biased autocorrelation sequence was computed using

$$\hat{r}(k) = \frac{1}{N} \sum_{n=0}^{N-k-1} y(n)y(n+k) \quad (6.6)$$

where N denotes the frame length.

The autocorrelation values were arranged to form the autocorrelation matrix required for solving the Yule–Walker equations.

6.5 Levinson Recursion

The Levinson recursion algorithm was applied to the autocorrelation sequence of each frame.

The algorithm recursively generated

- predictor coefficients,
- prediction-error energies,
- reflection coefficients.

The reflection coefficients obtained at the final stage served as the parameters of the lattice filter realization.

6.6 Analysis Lattice Filter

The analysis lattice filter was implemented using the reflection coefficients generated by the Levinson recursion algorithm.

For the m^{th} lattice stage,

$$e_m^+(n) = e_{m-1}^+(n) + k_m e_{m-1}^-(n-1) \quad (6.7)$$

$$e_m^-(n) = k_m e_{m-1}^+(n) + e_{m-1}^-(n-1). \quad (6.8)$$

The final forward prediction error obtained at the last stage was considered as the residual sequence.

The intermediate forward and backward prediction errors generated at each stage were stored for subsequent gapped-function analysis.

6.7 Synthesis Lattice Filter

The residual sequence produced by the analysis lattice was applied to the synthesis lattice filter.

Using the same reflection coefficients, the signal was reconstructed recursively through all lattice stages.

The reconstructed signal obtained at the output of the final stage was compared with the original speech signal to verify the correctness of the identified model.

6.8 Correlation Analysis

To study the decorrelation properties of the lattice filter, autocorrelation sequences were computed at various stages of the analysis and synthesis structures.

The correlation characteristics of

- the original speech signal,
- the residual sequence,
- the reconstructed signal

were compared.

The reduction of correlation in the residual sequence was used as an indicator of successful prediction and system identification.

6.9 Performance Evaluation

The performance of the identified model was evaluated by comparing the reconstructed speech signal with the original speech signal.

The reconstruction error was calculated as

$$e_r(n) = y(n) - \hat{y}(n) \quad (6.9)$$

where $y(n)$ denotes the original speech signal and $\hat{y}(n)$ denotes the reconstructed signal.

The mean-square error (MSE) was computed using

$$\text{MSE} = \frac{1}{N} \sum_{n=0}^{N-1} e_r^2(n). \quad (6.10)$$

A small reconstruction error indicates that the identified model accurately represents the original system.

6.10 Simulation Procedure

The complete simulation procedure may be summarized as follows:

1. Read the speech signal.
2. Divide the signal into frames.
3. Compute the autocorrelation sequence.
4. Apply Levinson recursion.
5. Extract reflection coefficients.
6. Implement the analysis lattice filter.
7. Obtain the residual sequence.
8. Implement the synthesis lattice filter.
9. Reconstruct the speech signal.
10. Evaluate the reconstruction error and correlation characteristics.

7 Results and Discussion

7.1 Introduction

This chapter presents the results obtained from the implementation of the linear prediction-based system identification algorithm. The speech signal was processed frame by frame using the autocorrelation method, Levinson recursion, and lattice filter realization. The predictor coefficients, reflection coefficients, gapped functions, correlation characteristics, and reconstruction performance were analyzed to evaluate the effectiveness of the identified model.

7.2 Predictor Coefficients

The predictor coefficients were obtained by solving the Yule–Walker equations using the Levinson recursion algorithm. These coefficients characterize the autoregressive model associated with each speech frame.

Prediction Coefficients for each Frame are listed below:

Frame 1:1.0000, 0.6272, 0.8785, 0.0471, -0.0562, -0.2214, -0.1470, 0.0854, 0.0396
Frame 2:1.0000, 0.4065, 0.8886, -0.2037, -0.2775, -0.3856, -0.5185, 0.0148, -0.1694
Frame 3:1.0000, -0.6184, 0.1991, -0.5071, 0.0734, 0.2261, -0.0419, 0.2403, -0.1301
Frame 4:1.0000, -0.8344, 0.0148, -0.0994, 0.1553, 0.0656, 0.0178, 0.0057, -0.0373
Frame 5:1.0000, -1.0798, 0.1448, -0.0197, 0.0739, 0.1638, 0.0408, -0.0924, 0.0706
Frame 6:1.0000, -1.6799, 0.7583, -0.1810, 0.1522, 0.3058, -0.2602, -0.0708, 0.0833
Frame 7:1.0000, -1.2531, 0.2558, -0.1066, 0.2782, 0.0863, 0.0008, -0.1597, 0.1219
Frame 8:1.0000, -1.1105, 0.2294, 0.0065, 0.0170, 0.0983, -0.0178, 0.0322, 0.0820

7.3 Reflection Coefficients

The reflection coefficients generated by the Levinson recursion algorithm constitute the model parameters of the lattice filter realization. These coefficients determine the behavior of each stage of the analysis and synthesis lattice structures.

The optimal reflection coefficients obtained using the Levinson recursion algorithm are listed below.

Frame 1: 0.3316, 0.7412, 0.0648, 0.2416, -0.1445, -0.2208, 0.0606, 0.0396
Frame 2: 0.3121, 0.8151, -0.1873, 0.2051, -0.3831, -0.4182, 0.0861, -0.1694
Frame 3: -0.5831, -0.0060, -0.3600, 0.2809, 0.1864, 0.0830, 0.1626, -0.1301
Frame 4: -0.7983, 0.0954, 0.0871, 0.2099, 0.0600, -0.0029, -0.0255, -0.0373
Frame 5: -0.9003, 0.4837, 0.2709, 0.2732, 0.1827, 0.0132, -0.0162, 0.0706
Frame 6: -0.9490, 0.8943, 0.1151, 0.1908, -0.0836, -0.2093, 0.0696, 0.0833
Frame 7: -0.9325, 0.6965, 0.3078, 0.3250, 0.0529, -0.0398, -0.0071, 0.1219
Frame 8: -0.9061, 0.4995, 0.2269, 0.2205, 0.1905, 0.1039, 0.1241, 0.0820

Since the lattice realization is completely specified by the reflection coefficients, these parameters represent the identified system in the present work.

7.4 Gapped Functions

The forward and backward prediction errors obtained at each lattice stage were analyzed to study the decorrelation process within the lattice structure.

It can be observed that the forward prediction error signals evolve progressively through the lattice stages. The final-stage residual differs significantly from the input signal, indicating that predictable signal components have been removed by the analysis lattice filter.

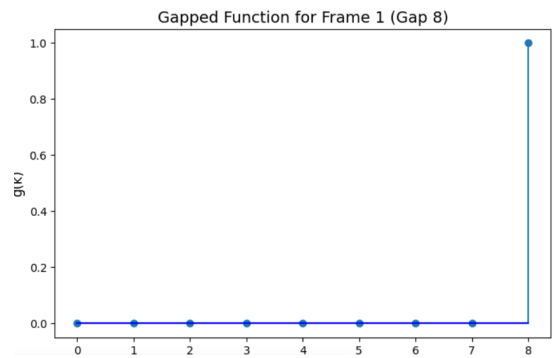
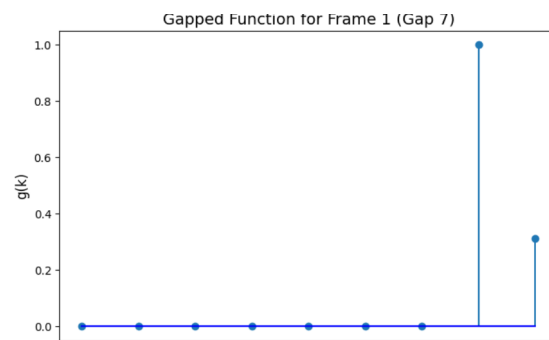
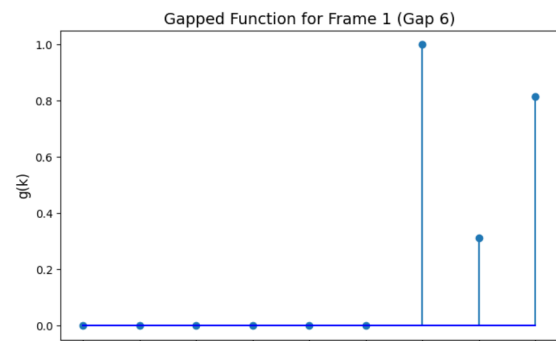
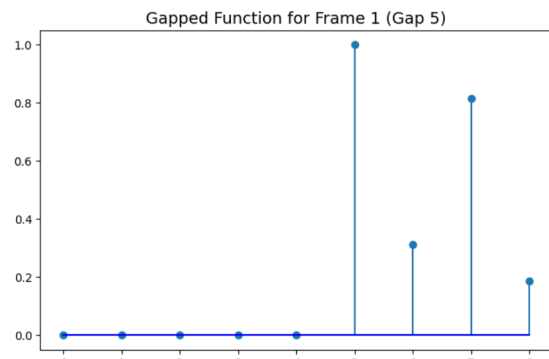
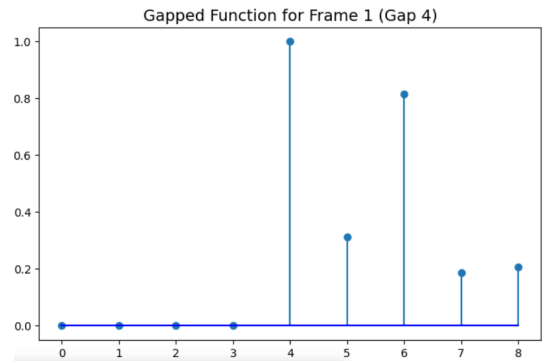
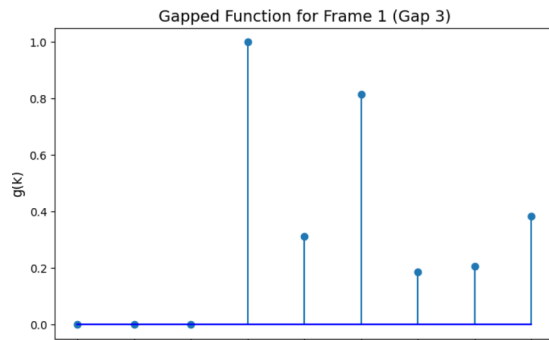
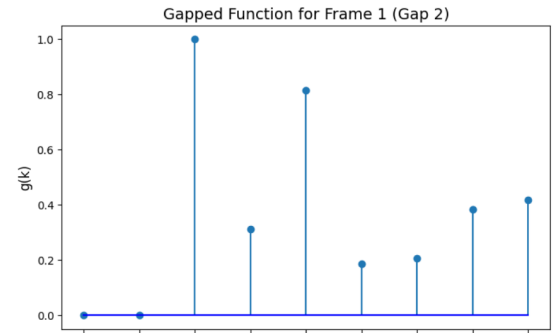
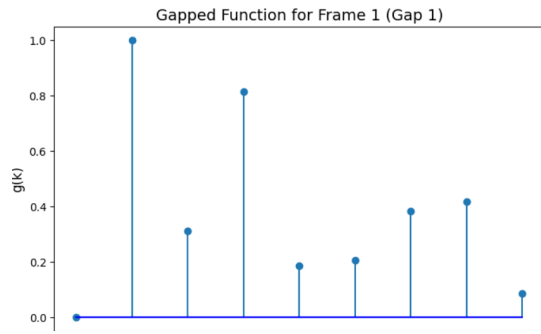


Figure 7.2: Gapp Function (Frame 3)

7.5 Analysis Lattice Correlation Study

The autocorrelation sequence of the original speech signal was compared with the autocorrelation sequence of the residual signal produced by the analysis lattice filter.

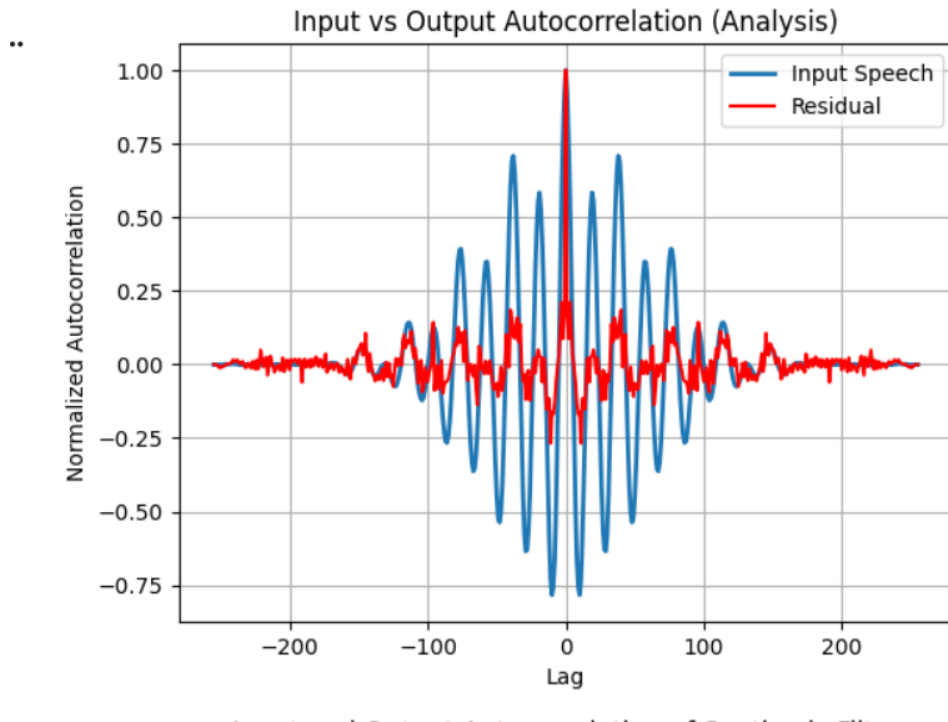


Figure 7.3: Normalized autocorrelation of the input speech signal and the residual signal obtained from the analysis lattice filter.

The residual signal exhibits significantly reduced correlation compared to the original speech signal. This demonstrates the ability of the analysis lattice filter to decorrelate the input sequence.

7.6 Synthesis Lattice Correlation Study

The synthesis lattice filter was used to reconstruct the speech signal from the residual sequence. The autocorrelation characteristics of the reconstructed signal were compared with those of the residual sequence.

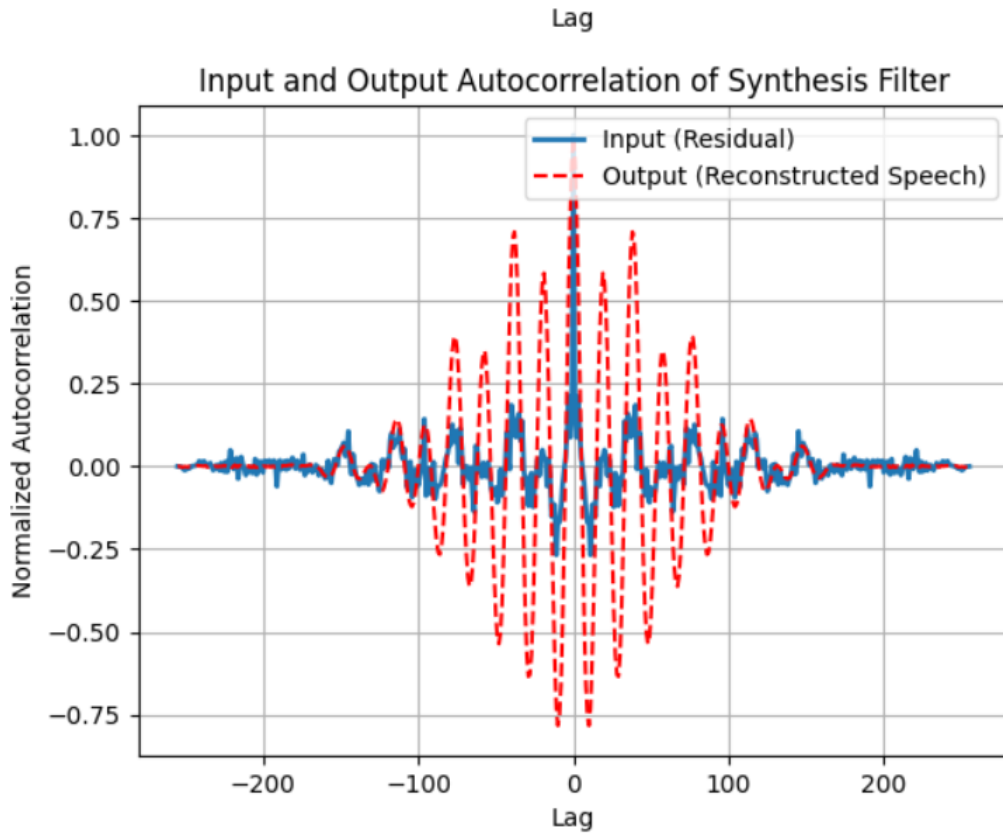


Figure 7.4: Normalized Autocorrelation Before and After Synthesis

The autocorrelation structure of the reconstructed signal closely matches that of the original speech signal, confirming the effectiveness of the synthesis lattice realization.

7.7 Signal Reconstruction

The reconstructed speech signal obtained from the synthesis lattice filter was compared with the original speech signal.

Figure 7.5 presents the original and reconstructed waveforms.

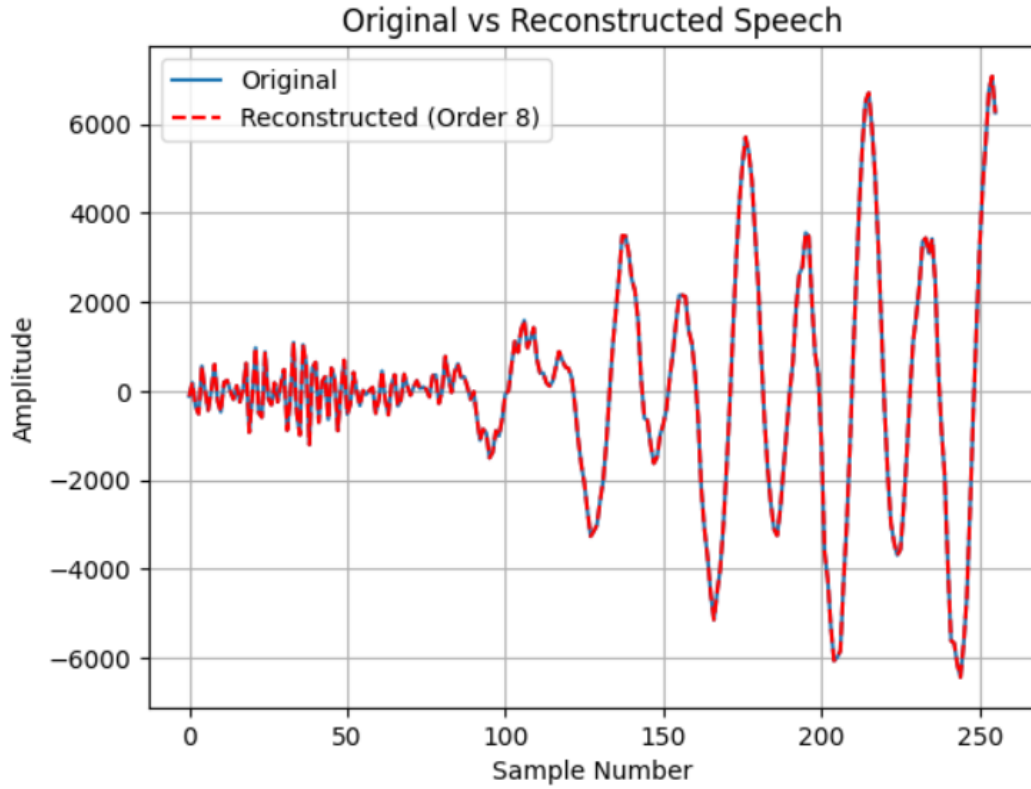


Figure 7.5: Original and Reconstructed Speech Signals

A close agreement between the two waveforms indicates that the identified model successfully captures the characteristics of the original system.

7.8 Reconstruction Error

The reconstruction error was computed as

$$e_r(n) = y(n) - \hat{y}(n) \quad (7.1)$$

where $y(n)$ is the original signal and $\hat{y}(n)$ is the reconstructed signal.

The Mean Square Error (MSE) obtained from the simulation was

$$\text{MSE} = 1.327635 \quad (7.2)$$

A low MSE value indicates accurate reconstruction and validates the identified system model.

7.9 Discussion

The simulation results demonstrate the effectiveness of linear prediction for system identification. The predictor coefficients obtained from the Yule–Walker equations successfully modeled the speech signal, while the reflection coefficients provided an efficient lattice realization.

The analysis lattice filter progressively decorrelated the speech signal, as evidenced by the reduction in autocorrelation of the residual sequence. The synthesis lattice filter reconstructed the original signal with minimal distortion, resulting in a low reconstruction error.

The gapped functions obtained at intermediate lattice stages provided additional insight into the decorrelation process. The results confirm that the combination of the autocorrelation method, Levinson recursion, and lattice filter realization constitutes an effective framework for system identification and signal modeling.

8 Conclusions

8.1 Conclusions

In this work, an unknown linear system was identified using linear prediction techniques. The speech signal was analyzed on a frame-by-frame basis, and the autocorrelation method was employed to estimate the signal correlation structure. The Yule–Walker equations were formulated and solved using the Levinson recursion algorithm to obtain the optimal predictor coefficients.

The Levinson recursion algorithm not only provided an efficient solution to the Yule–Walker equations but also generated the reflection coefficients required for lattice filter realization. These reflection coefficients were used as the model parameters in the implementation of the analysis and synthesis lattice filters.

The analysis lattice filter successfully transformed the correlated speech signal into a prediction residual with reduced correlation. The intermediate forward and backward prediction errors, referred to as gapped functions, were examined to study the decorrelation process at various lattice stages. The synthesis lattice filter reconstructed the original speech signal from the residual sequence using the same set of reflection coefficients.

The simulation results demonstrated that the reconstructed signal closely matched the original speech signal, indicating successful system identification and model realization. The correlation studies further confirmed the decorrelation capability of the analysis lattice and the reconstruction capability of the synthesis lattice. The low reconstruction error obtained from the simulations validates the effectiveness of the proposed approach.

Overall, the combination of the autocorrelation method, Levinson recursion algorithm, and lattice filter realization provides an efficient and reliable framework for system identification and signal modeling.

Bibliography

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